

Lie Theory for the Impatient

Nir Elber

Fall 2026

Abstract

This document collects a variety of definitions and results from Lie theory. To avoid distraction, all vector spaces are over $\mathbb{C}$, and all manifolds are real.

Contents

Contents	1
1 Lie Algebras	2
1.1 Starting Out	2
1.2 Nilpotent and Solvable Lie Algebras	3
1.3 Simple and Semisimple Lie Algebras	5
1.4 The Root Decomposition	7
1.5 Representation Theory of Semisimple Lie Algebras	8

1 Lie Algebras

1.1 Starting Out

Definition 1 (Lie algebra). A Lie algebra is a vector space $\mathfrak{g}$ equipped with a Lie bracket $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the following properties.

- (i) Bilinear: the map $[-, -]$ is bilinear.
- (ii) Alternating: $[X, X] = 0$ for all $X \in \mathfrak{g}$.
- (iii) Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

A morphism of Lie algebras is a linear map preserving the bracket.

Example 2. For any (possibly non-commutative, non-unital) algebra A over $\mathbb{C}$, one can define the Lie bracket $[-, -]: A \times A \rightarrow A$ by

$$[a, b] := ab - ba.$$

For example, applying this to $A = \text{End}(V)$ where V is a vector space gives the Lie algebra $\mathfrak{gl}(V)$. If further $V = \mathbb{C}^n$, we abbreviate $\mathfrak{gl}(V)$ to $\mathfrak{gl}_n(\mathbb{C})$.

Example 3. The subspace

$$\mathfrak{sl}_n(\mathbb{C}) := \{X \in \mathfrak{gl}_n(\mathbb{C}) : \text{tr } X = 0\}$$

is a Lie subalgebra.

Example 4. The subspace of alternating matrices is a Lie subalgebra, called $\mathfrak{o}_n(\mathbb{C}) \subseteq \mathfrak{gl}_n(\mathbb{C})$.

Definition 5 (representation). A representation of a Lie algebra $\mathfrak{g}$ is a vector space V along with a morphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. A morphism of representations is a morphism of the underlying vector spaces which preserves the $\mathfrak{g}$ -action.

Example 6 (adjoint). The adjoint representation of a Lie algebra $\mathfrak{g}$ is the morphism $\text{ad}_\bullet: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ defined by

$$\text{ad}_X(Y) := [X, Y].$$

The fact that $\text{ad}_\bullet$ preserves the Lie bracket is equivalent to the Jacobi identity.

Definition 7 (irreducible). A representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is *irreducible* if and only if $V \neq 0$, and V admits no nonzero proper subrepresentations. A representation is *completely reducible* if and only if it is a direct sum of irreducible representations.

Example 8. Let $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the Lie subalgebra of upper-triangular matrices. Then the representation $\mathfrak{b} \rightarrow \mathfrak{gl}_2(\mathbb{C})$ is not completely reducible.

Lemma 9 (Schur). Let $\rho_V: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ and $\rho_W: \mathfrak{g} \rightarrow \mathfrak{gl}(W)$ be representations, and let $\varphi: V \rightarrow W$ be a nonzero morphism of representations.

- (a) If V is irreducible, then φ is injective.
- (b) If W is irreducible, then φ is surjective.
- (c) If both V and W are irreducible, then φ is an isomorphism.

Sketch. Here, (c) follows from (a) and (b). We only sketch (a): because φ is nonzero, $\ker \varphi$ is a proper subrepresentation of V , so $\ker \varphi = 0$ because V is irreducible. ■

Example 10. Let the natural action of $\mathfrak{sl}_2(\mathbb{C})$ on $\mathbb{C}^2$ is irreducible. It turns out that the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$ are exactly the symmetric powers of $\mathbb{C}^2$. To see this, show that $\text{diag}(+1, -1)$ acts semisimply on any irreducible representation, and then one can parameterize the irreducible representations by the highest eigenvalue of $\text{diag}(+1, -1)$.

Definition 11 (universal enveloping algebra). Fix a Lie algebra $\mathfrak{g}$. Then the *universal enveloping algebra* $U\mathfrak{g}$ of $\mathfrak{g}$ is the free complex algebra generated by $\mathfrak{g}$ along with the relations

$$XY - YX = [X, Y]$$

for each $X, Y \in \mathfrak{g}$.

Remark 12. The data of a representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is equivalent to the data of a ring homomorphism $U\mathfrak{g} \rightarrow \text{End}(V)$.

Theorem 13 (Poincaré–Birkhoff–Witt). Fix a Lie algebra $\mathfrak{g}$, and let $\{X_i\}_{i=1}^n$ be an (ordered) basis. Then the monomials

$$\{X_{i_1} \cdots X_{i_m} : i_1 \leq i_2 \leq \cdots \leq i_m\}$$

form a basis of $U\mathfrak{g}$.

Sketch. Certainly $U\mathfrak{g}$ is spanned by monomials, and one can use the relations $XY = YX + [X, Y]$ in order to rearrange the entries of the monomial. Thus, the given monomials span. Checking that they are linearly independent amounts to a long and technical argument. ■

1.2 Nilpotent and Solvable Lie Algebras

Definition 14 (ideal). Fix a Lie algebra $\mathfrak{g}$. Then an *ideal* $I \subseteq \mathfrak{g}$ is a subspace such that $[X, I] \subseteq I$ for any $X \in \mathfrak{g}$. Equivalently, I is a $\mathfrak{g}$ -subrepresentation of $\mathfrak{g}$.

Example 15 (center). Fix a Lie algebra $\mathfrak{g}$. Then the *center* is the subspace

$$\mathfrak{z}(\mathfrak{g}) := \{X \in \mathfrak{g} : [X, \mathfrak{g}] = 0\}.$$

The center is an ideal. It is kernel the adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$.

Example 16 (commutator). Fix a Lie algebra $\mathfrak{g}$. Then the *commutator* is the subspace $[\mathfrak{g}, \mathfrak{g}]$. By the Jacobi identity, $[\mathfrak{g}, \mathfrak{g}]$ is a Lie ideal.

Remark 17. If I is an ideal of a Lie algebra $\mathfrak{g}$, the quotient space $\mathfrak{g}/I$ is a Lie algebra, and there is a short exact sequence

$$0 \rightarrow I \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/I \rightarrow 0$$

of $\mathfrak{g}$ -modules.

Definition 18. A Lie algebra $\mathfrak{g}$ is *abelian* if and only if $[\mathfrak{g}, \mathfrak{g}] = 0$.

Example 19. For any Lie algebra, the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian.

Definition 20 (nilpotent). A Lie algebra $\mathfrak{g}$ is *nilpotent* if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that $\mathfrak{a}_{i+1}$ is an ideal of $\mathfrak{a}_i$ with abelian quotient.

Remark 21. It is equivalent to check if the sequence $\mathfrak{a}_{i+1} := [\mathfrak{g}, \mathfrak{a}_i]$ eventually stabilizes at 0.

Example 22. The subspace of strictly upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is nilpotent.

Definition 23 (solvable). A Lie algebra $\mathfrak{g}$ is *solvable* if and only if there is a sequence

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_k = 0$$

such that each $\mathfrak{a}_i$ is an ideal of $\mathfrak{g}$ such that $\mathfrak{g}/\mathfrak{a}_i$ is abelian.

Remark 24. It is equivalent to check if the sequence $\mathfrak{a}_i := [\mathfrak{a}_{i+1}, \mathfrak{a}_{i+1}]$ eventually stabilizes at 0.

Example 25. Any nilpotent Lie algebra is solvable.

Example 26. The subspace of upper-triangular matrices of $\mathfrak{gl}_n(\mathbb{C})$ is solvable but not nilpotent.

Theorem 27 (Lie). Let $\mathfrak{g}$ be a solvable Lie algebra. For any representation V of $\mathfrak{g}$, there is a basis in which $\mathfrak{g}$ acts by upper-triangular matrices.

Sketch. By induction on $\dim V$, it is enough to show that the action of $\mathfrak{g}$ on V admits a nonzero eigenvector. For this, we induct on $\dim \mathfrak{g}$. By solvability, one may find a codimension-1 ideal $\mathfrak{g}' \subseteq \mathfrak{g}$; choose any $X \in \mathfrak{g} \setminus \mathfrak{g}'$. Now, induction tells us that V admits a nonzero eigenvector v for $\mathfrak{g}'$. With some difficulty, can show that

$$W := \text{span}_{\mathbb{C}} X^{\mathbb{N}} v$$

is a $\mathfrak{g}$ -subrepresentation of V , and it turns out that $\mathfrak{g}'$ acts diagonally on all of W . Thus, any eigenvector of X acting on W will do. ■

Corollary 28. Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

Sketch. If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then one can show that $\mathfrak{g}$ is solvable by building ideals from the short exact sequence

$$0 \rightarrow [\mathfrak{g}, \mathfrak{g}] \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \rightarrow 0.$$

Conversely, if $\mathfrak{g}$ is solvable, then by Theorem 27, one can find a basis of $\mathfrak{g}$ on which the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ is given by upper-triangular matrices. Thus, the action of $[\mathfrak{g}, \mathfrak{g}]$ is given by strictly upper-triangular matrices. This is enough to show $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. ■

Remark 29. Similarly, if $\mathfrak{g}$ is nilpotent, one can show that any representation V admits a basis on which $\mathfrak{g}$ acts by strictly upper-triangular matrices.

Theorem 30 (Engel). Let $\mathfrak{g}$ be a Lie algebra. Then $\mathfrak{g}$ is nilpotent if and only if the operator $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent for all $X \in \mathfrak{g}$.

Sketch. If ad_X is nilpotent for all $X \in \mathfrak{g}$, then $[\mathfrak{g}, [\mathfrak{g}, [\mathfrak{g}, \dots, [\mathfrak{g}, \mathfrak{g}]]]] = 0$ for a high enough nesting level, so $\mathfrak{g}$ is nilpotent. Conversely, if $\mathfrak{g}$ is nilpotent, then Remark 29 grants a basis of $\mathfrak{g}$ upon which $\mathfrak{g}$ acts by strictly upper-triangular matrices, so ad_X is nilpotent for all $X \in \mathfrak{g}$. ■

1.3 Simple and Semisimple Lie Algebras

Definition 31 (simple). A Lie algebra $\mathfrak{g}$ is *simple* if and only if it is not abelian and admits no nonzero proper ideals.

Example 32. One can check by hand that $\mathfrak{sl}_2(\mathbb{C})$ is a simple Lie algebra.

Definition 33 (semisimple). A Lie algebra $\mathfrak{g}$ is *semisimple* if and only if it admits no nonzero solvable ideals.

Example 34. Any simple Lie algebra is semisimple. We will have many more examples shortly.

Remark 35. It turns out that any semisimple Lie algebra $\mathfrak{g}$ is a direct sum of simple ones. In short, one can decompose the adjoint action of $\mathfrak{g}$ on $\mathfrak{g}$ into a sum of irreducible subrepresentations, and each irreducible subrepresentation is a simple Lie subalgebra.

Remark 36. If $\mathfrak{g}$ is semisimple, then $\mathfrak{z}(\mathfrak{g}) = 0$, so the adjoint representation $\mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is injective.

Definition 37 (reductive). A Lie algebra $\mathfrak{g}$ is *reductive* if and only if $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is semisimple.

Example 38. Because $\mathfrak{z}(\mathfrak{gl}_2(\mathbb{C}))$ consists of the scalar matrices, the Lie algebra $\mathfrak{gl}_2(\mathbb{C})$ is reductive.

Remark 39. A reductive Lie algebra $\mathfrak{g}$ is semisimple if and only if $\mathfrak{z}(\mathfrak{g}) = 0$.

Definition 40 (radical). Fix a Lie algebra $\mathfrak{g}$. Then the *radical* $\text{rad } \mathfrak{g}$ is the sum of all solvable ideals of $\mathfrak{g}$. One can check that $\text{rad } \mathfrak{g}$ is a solvable ideal.

Example 41. For a Lie algebra $\mathfrak{g}$, we see $\text{rad } \mathfrak{g} = 0$ if and only if $\mathfrak{g}$ is semisimple. For example, $\mathfrak{g}/\text{rad } \mathfrak{g}$ is semisimple.

Theorem 42 (Levi decomposition). Fix any Lie algebra $\mathfrak{g}$. Then there is a semisimple Lie subalgebra $\mathfrak{g}_{\text{ss}} \subseteq \mathfrak{g}$ and a direct sum decomposition

$$\mathfrak{g} = \text{rad } \mathfrak{g} \oplus \mathfrak{g}_{\text{ss}}.$$

Sketch. This is a very difficult result. The idea is to show that the short exact sequence

$$0 \rightarrow \text{rad } \mathfrak{g} \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}_{\text{ss}} \rightarrow 0$$

admits a splitting. Such an extension problem reduces to a calculation of some cohomology groups, which is more or less done by hand. ■

Proposition 43. Let V be a representation of a Lie algebra $\mathfrak{g}$. Suppose that the pairing $\langle -, - \rangle_V : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ defined by

$$\langle X, Y \rangle_V := \text{tr}(XY; V)$$

is non-degenerate. Then $\mathfrak{g}$ is reductive.

Sketch. It is enough to show that any $X \in [\mathfrak{g}, \text{rad } \mathfrak{g}]$ lives in the kernel of $\langle -, - \rangle_V$ (and thus vanishes). By passing to some short exact sequences, we may assume that V is irreducible. Using Theorem 27, one can show that each $\text{rad}(\mathfrak{g})$ -eigenspace of V is $\mathfrak{g}$ -invariant, so irreducibility implies that $\text{rad } \mathfrak{g}$ acts by scalars on V . A direct calculation now shows that $[\mathfrak{g}, \text{rad } \mathfrak{g}]$ acts by zero on V . ■

Example 44. The standard representation $V = \mathbb{C}^n$ of $\mathfrak{gl}_n(\mathbb{C})$ satisfies the hypothesis of Proposition 43, so $\mathfrak{gl}_n(\mathbb{C})$ is reductive. A similar argument holds for $\mathfrak{so}_n(\mathbb{C})$.

Theorem 45 (Cartan). Let $\mathfrak{g}$ be a semisimple Lie algebra. Then the pairing

$$\langle X, Y \rangle_{\mathfrak{g}} := \text{tr}(XY; \mathfrak{g})$$

is non-degenerate. This pairing is called the *Killing form*.

Sketch. Please skip this sketch. Let $\mathfrak{h}$ be the kernel of $\langle -, - \rangle_{\mathfrak{g}}$, which is an ideal. We will show that $\mathfrak{h}$ is a solvable, which forces $\mathfrak{h} = 0$. It is enough to show that $\text{ad}(\mathfrak{h}) \subseteq \mathfrak{gl}(\mathfrak{h})$ is solvable. Some linear algebra shows that the pairing $\langle X, Y \rangle := \text{tr}(XY; \mathfrak{h})$ on $\mathfrak{h}$ vanishes. Thus, we are in the situation that we have a Lie algebra $\mathfrak{h}'$ with a faithful representation V for which

$$\text{tr}(XY; V) = 0$$

for each $X, Y \in \mathfrak{h}'$. One can use the Jordan decomposition to show that any element of $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $[\mathfrak{h}', \mathfrak{h}']$ is nilpotent, so $\mathfrak{h}'$ is solvable. ■

1.4 The Root Decomposition

Definition 46. Fix a Lie algebra $\mathfrak{g}$. Then $X \in \mathfrak{g}$ is *nilpotent* or *semisimple* if and only if $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is nilpotent or semisimple, respectively.

Proposition 47 (Jordan decomposition). Fix a semisimple Lie algebra $\mathfrak{g}$. For any $X \in \mathfrak{g}$, there is a unique semisimple $X_s \in \mathfrak{g}$ and $X_n \in \mathfrak{g}$ such that $[X_s, X_n] = 0$ and $X = X_s + X_n$.

Sketch. The usual Jordan decomposition produces a unique pair $(X'_s, X'_n) \in \mathfrak{gl}(\mathfrak{g})^2$ for which X'_s is semisimple, X'_n is nilpotent, $[X'_s, X'_n] = 0$, and $\text{ad}_X = X'_s + X'_n$. Thus, it remains to check that X'_s and X'_n are in the image of $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. In short, the Jordan decomposition shows that X'_s and X'_n are polynomials in ad_X , so they are derivations of $\mathfrak{g}$, but all derivations of $\mathfrak{g}$ come from $\mathfrak{g}$. ■

Definition 48 (Cartan). Fix a semisimple Lie algebra $\mathfrak{g}$. Then a *torus* is an abelian subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ consisting of semisimple elements. A maximal torus is a *Cartan subalgebra*.

Example 49. The subalgebra of diagonal matrices of $\mathfrak{sl}_n(\mathbb{C})$ is a Cartan subalgebra.

Remark 50. Let $\mathfrak{h}$ be a Cartan subalgebra. It turns out that

$$\mathfrak{h} = \{X \in \mathfrak{h} : [X, \mathfrak{h}] = 0\}.$$

In short, use the Jordan decomposition and the maximality of $\mathfrak{h}$.

Remark 51. It turns out that any two Cartan subalgebras of $\mathfrak{g}$ are conjugate under $\text{Aut } \mathfrak{g}$. This is fairly technical, and we will only use it to show that certain constructions are independent of the choice of $\mathfrak{h}$.

Definition 52 (rank). Fix a semisimple Lie algebra $\mathfrak{g}$. Then the *rank* of $\mathfrak{g}$ is the dimension of a Cartan subalgebra of $\mathfrak{g}$.

Example 53. The rank of $\mathfrak{sl}_n(\mathbb{C})$ is $n - 1$.

Definition 54. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root system* $\Phi \subseteq \mathfrak{h}^*$ is the set of nonzero eigenspaces of $\mathfrak{h}$ acting on $\mathfrak{g}$.

Proposition 55. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then the *root decomposition* is the eigenspace decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha},$$

where $\mathfrak{g}_{\alpha}$ is the eigenspace of $\alpha \in \Phi$.

Sketch. Diagonalize the representation $\mathfrak{g}$ of $\mathfrak{h}$. ■

Remark 56. For $\alpha, \beta \in \Phi$, we have $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$.

Remark 57. The subspace Φ spans $\mathfrak{h}^*$. Indeed, any $h \in \mathfrak{h}$ orthogonal to all $\alpha \in \Phi$ then has $\text{ad}_h = 0$, so $h \in \mathfrak{z}(\mathfrak{g})$, so $h = 0$.

Notation 58. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. For any $\alpha \in \Phi$, the subspaces $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ generate a subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2(\mathbb{C})$. (This is a little tricky to show.) We denote this subalgebra by $\mathfrak{sl}_2(\mathbb{C})_\alpha$.

Proposition 59. Fix a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ and root system $\Phi \subseteq \mathfrak{h}^*$. Let $(-, -)$ be the Killing form on $\mathfrak{g}$.

- (a) For each $\alpha \in \Phi$, $\dim \mathfrak{g}_\alpha = 1$.
- (b) For each $\alpha, \beta \in \Phi$, $2(\alpha, \beta)/(\alpha, \alpha) \in \mathbb{Z}$.
- (c) For each $\alpha \in \Phi$, define the reflection $s_\alpha : \mathfrak{h}^* \rightarrow \mathfrak{h}^*$ defined by $s_\alpha(\lambda) := \lambda - 2(\alpha, \lambda)/(\alpha, \alpha) \cdot \alpha$. Then s_α preserves Φ .
- (d) If $\alpha \in \Phi$ and $k\alpha \in \Phi$ for some $k \in \mathbb{C}$, then $k \in \{+1, -1\}$.

Sketch. Let $(-, -)$ be the Killing form on $\mathfrak{h}^*$, which is non-degenerate. Define $h_\alpha := 2H_\alpha/(\alpha, \alpha)$, where H_α is the element dual to $\alpha \in \mathfrak{h}^*$. Then one can show that

$$\mathbb{C}h_\alpha \oplus \bigoplus_{k \in \mathbb{Z} \setminus \{0\}} \mathfrak{g}_{k\alpha}$$

is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})_\alpha$. Each part now follows from understanding the irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$. ■

1.5 Representation Theory of Semisimple Lie Algebras

Theorem 60. Let $\mathfrak{g}$ be a semisimple Lie algebra. Then all representations of $\mathfrak{g}$ are completely reducible.

Sketch. There are many proofs of this theorem. The main idea is to show that any short exact sequence

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$$

of representations of $\mathfrak{g}$ splits as a direct sum $V = V' \oplus V''$. The most algebraic proof proceeds by finding “Casimir” elements in the center of $U\mathfrak{g}$. By Lemma 9, they act by scalars on irreducible representations. One can force the splitting if granted a Casimir elements which acts on V' and V'' by different scalars. ■